Lesson Planning

Complete Lesson Plan Book

Class: Class 11

Subject: Computer Science

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Binary Number System: Converting Between Binary and Decimal

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Learning Objectives

• Students will explain why computers use the binary number system.

• Students will convert binary numbers to decimal using place values.

• Students will convert decimal numbers to binary by repeated division by 2.

Key Concepts

• Computers work with two states, ON and OFF, so they store and process data as binary digits, 0 and 1.

• In binary, each place value is a power of 2: from the right, 1, 2, 4, 8, 16 and so on.

• To convert binary to decimal, multiply each bit by its place value and add the results; for example 1011 = 8 + 0 + 2 + 1 = 11.

• To convert decimal to binary, divide repeatedly by 2 and read the remainders from bottom to top.

Teaching Methodology (Step by Step)

1. Ask why a light switch has only two states, and link that to 0 and 1 in a computer.
2. Build the binary place-value table on the board and convert 1101 together.
3. Model decimal-to-binary conversion of 13 by repeated division by 2.
4. Students convert four numbers in each direction in pairs.
5. Students check each other's answers by converting back.

Classroom Activities

• Place-value race: pairs convert binary numbers against the clock.

• Remainder chain: students build the division chain for a decimal number on the board.

• Back-and-forth check: students convert a number one way and a partner converts it back.

Assessment Questions

1. Convert 10101_2 to decimal.
2. Convert 19 to binary by repeated division by 2.
3. Why do computers use binary instead of decimal?

Homework

• Convert 11010_2 and 100111_2 to decimal.

• Convert 25 and 42 to binary, and check each answer by converting back.

Time Allocation

40 minutes